

Max T. Pan

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Education

Georgia Institute of Technology | Atlanta, GA

PhD in Electrical and Computer Engineering

August 2022 – Present

MS in Electrical and Computer Engineering | GPA: 3.69

August 2022 – May 2025

University of California, Berkeley | Berkeley, CA

BA in Physics, BA in Computer Science | GPA: 3.85

August 2018 – May 2022

Employment

Georgia Tech Dynamic Robotic Adaptive Technologies (DART) Lab

Atlanta, GA

Graduate Research Assistant

May 2025 - Present

- Benchmarked just-in-time vs. ahead-of-time motion planning algorithms in Julia on the single-board computers for real-time onboard planning applications.
- Implemented an object detector, utilizing the light-head R-CNN + region-of-interest (RoI) transformer architecture and trained on satellite images, for a landmark navigation project using the AirSim simulation environment. Inference time optimizations for the object detector were explored using TensorRT.
- Explored explainable methods for long short-term memory (LSTM) models with the application towards motion planning

Georgia Tech College of Computing

Atlanta, GA

Graduate Teaching Assistant for “CS 7641 Machine Learning”

January 2025 – May 2025

- Collaborated with a teaching team to assist 800+ students
- Graded assignments reports, held weekly office hours, gave lectures on supervised & unsupervised machine learning techniques

Molecular Foundry at Lawrence Berkeley National Lab (LBNL)

Berkeley, CA

Undergraduate Researcher

September 2020 – May 2022

- Performed Density Functional Theory (DFT) calculations to look for Metal-Organic Frameworks (MOFs) that could perform nitrogen backbonding, which have strong potential in nitrogen & methane separation and thus natural gas purification

Personal Projects

- Implemented Unconditional Latent Diffusion Model (LDM) to generate images using Vector Quantized Variational Autoencoder (VQ-VAE), cosine beta scheduling, and Denoising Diffusion Implicit Models (DDIM). CelebA-HQ dataset was used.
- Designed, implemented, and trained Deep Convolutional Generative Adversarial Network (DCGAN) and Denoising Diffusion Probabilistic Model (DDPM) to generate images. Yu-Gi-Oh! card and FGVC Aircraft dataset were used to compare/contrast the effectiveness of two models. FID scores were calculated.
- Implemented Vision Transformer (ViT) for image classification, trained on CIFAR100 dataset.
- Designed and implemented a Galaxian-playing agent using Double Q-Learning and OpenAI Gym
- Implemented Contrastive Language-Image Pre-training (CLIP) model, trained on the flickr8k dataset, along with applications in text-to-image search, image-to-image search, and zero-shot classification
- Implemented RRT* algorithm in C++ for collision-free path planning in a 3D obstacle environment
- Implemented and benchmarked naïve, tiled, and shared-memory matrix multiplication algorithms in CUDA
- Implemented a Jacobian-transpose Cartesian PID controller with gravity compensation for a simulated PUMA robotic manipulator in Simulink/Simscape, enabling end-effector trajectory tracking for symbol and character drawing tasks and achieving sub-millimeter trajectory tracking accuracy (~0.25 mm RMSE)

Publications

- L. Gustin, L. Xing, **Max T. Pan**, R. Jin, W. Xie, “Electron Counts, Structural Stability, and Magnetism in BaCuSn₂-Type YM_xGe₂ (M = Cr, Mn, Fe, Co, and Mn)”, *J. Alloys & Compounds* **741**, 840 (2018).
- J. Brooks, G. L. Knapp, J. Yuan, C. G. Lowery, **Max T. Pan**, B. E. Cadigan, S. Guo, D. S. Hussey, L. G. Butler, “Neutron Imaging of Laser Melted SS316 Test Objects with Spatially Resolved Small Angle Neutron Scattering”, *J. Imaging* **3**, 58 (2017).
- X. Gui, T. R. Chang, T. Kong, **Max T. Pan**, R. J. Cava, W. Xie, “Monoclinic 122-Type BaIr₂Ge₂ with a Channel Framework: A Structural Connection between Clathrate and Layered Compounds”, *Materials* **10**, 818 (2017).

Skills

- **Programming Languages:** Python, Java, C, C++, CUDA, SQL, LaTeX, MatLab, LabView, Julia
- **Software & Tools:** Git, Pandas, NumPy, Matplotlib, Seaborn, PyTorch, Sci-kit Learn, OpenMMLab, Hugging Face, OpenCV, MuJoCo, Simulink

Relevant Coursework

- **Computer Science:** Data Structures, Efficient Algorithms & Intractable Problems, Great Ideas in Computer Architecture
- **AI/ML/Robotics:** Introduction to Artificial Intelligence, Machine Learning, Natural Language Processing, Computer Vision, Graphical Models in ML, Robotics